bpm-linearity guide

Wenu-imaging

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Executive summary

For each detector, load calibrations and raw planes once, capture raw evidence, scan-correct, trim, subtract the supplied active master from flats and retain one processed cube. If an input BPM is supplied, log the skip; otherwise create a robust response BPM. Measure linearity ROIs, pairs and direct sites from that same cube, publish FITS/ASDF, then release it. Linearity is measurement only: no physical response correction, shutter solution or gain conversion is applied.

Qualification. Earlier CCD65 saved-product checks passed. The classified-list interface still requires Mac qualification; WFI and full-mosaic review are pending.

Command and help

Run from orchestration with a fresh empty reduced run:

```
mefpp_recipe cal bpm-linearity --config RECIPE.toml --qc false
```

```
mefpp --help
mefpp_recipe cal --help
mefpp_recipe cal bpm-linearity --help
mefpp_recipe cal linearity --help
mefpp_recipe cal bias --help
```

Help is data-free: categories list recipes and pending entries; operation help reports inputs, options, defaults and outputs.

Public option	Meaning / default
`--config FILE`	Required recipe TOML; references project/settings/inputs.
`--qc true|false`	Optional override; CLI > recipe `qc` > true.
`--help`	Context-specific help; successful exit, no processing.

Inputs and directory contract

Project TOML defines raw, calibration, reduced, work and log roots; its containing directory is orchestration. Raw observations and input calibrations are read-only. Lists/TOMLs belong in orchestration; products, work and logs belong below reduced. Recipe `inputs.list` references a separate ordered text file:

```
OMEGA.frame1.fits  FLAT,DOME
OMEGA.frame2.fits  FLAT,DOME
```

Each row is path plus `DPR.TYPE`; tabs permit spaces in paths. Comments and blank lines are ignored. Image/list paths use the raw/recipe roots respectively. Invalid or duplicate inputs fail. Save order, classifications and bindings.

Supply a reduced master bias and optional BPM; dependency-list reduction is deferred. Optional BIAS rows provide diagnostics, never master construction. Flat pairs derive from classification and header exposure times: exactly two members per group within the profile tolerance; missing/ambiguous partners fail. Review calibration, timing tolerance and site before using the WFI example.

Recipe settings and defaults

Templates under `config/recipes/`:

`omegacam-bpm-linearity.example.toml`, `wfi-bpm-linearity.example.toml`. They initially select one detector; extend in detector-catalogue order after reviewing the actual bindings.

Field	Default / requirement
`project`, `settings`, `run`	Required project/native settings; fresh run relative to reduced.
`qc`	true; optional diagnostics enabled.
`assembly.reference_detector`	Required selected detector; among created BPMs if any.
`preparation.detector_toml`	Required detector identity/geometry file.
`preparation.processing_toml`	Required scan/trim/ROI and measurement controls.
`bpm.acquisition`	Required: `quick` or `provisional`.
`bpm.output_mode`	Required: `new` or `combined`.
`bpm.response_low/high`	0.5 / 1.5; finite, $0 <$ low $<$ high.
`bpm.upper_fraction`	Required, $(0, 1]$; example 0.85.
`bpm.minimum_span_fraction`	Required, $(0, 1]$; example 0.15.
`bpm.maximum_pair_difference`	Required, ≥ 0; example 0.02.
`inputs.list`	Classified observation list; no inline paths or pair IDs.
`detectors`	Explicit detector/configuration/HDU/readout bindings.

Each detector declares `master_bias`, `master_hdu`, `master_sha256`, reference ROI IDs, direct `sites` (ROI, active x/y), and pixel/row bounds. `input_bpm` is an explicit calibration path with `bpm_hdu`, or `omitted`. Common `hdu` is required; optional `flat_hdus`/`bias_hdus` arrays override it per input. Raw paths use the raw root; master/BPM paths use its calibration root. Original indices are list-row ordinals. Rectangular ROIs are defined in the processing profile (`xmin/xmax/ymin/ymax`); named direct sites use single-pixel `x/y`. The two are distinct measurements.

Optional `identity` defaults: `identity_key=EXTNAME`, `amplifier_key=-` (unused), bin keys `HIERARCH ESO DET WIN1 BINX/BINY`, `exposure_key=EXPTIME`, `exposure_hdu=0`. Current profiles require 1x1 binning.

Processing TOML explicitly supplies scan switches, sampled scan rectangles, low/high rejection counts, retained rectangles, ROI rectangles, saturation, mask encoding/polarity/reject bits, pair-time tolerance, bias-admission/stack controls and `with_mad`. These have no silent instrument defaults. OCAM examples use saturation 65534 ADU, rejection 10/10, bias range 0–1000 ADU, stack kappa 2, minimum 3 frames, binary good value 1 and exact pair-time matching. These are configured values, not universal detector properties.

`qc=false` skips optional BPM full-frame QC and MAD, retaining requested ROI means, variances, populations, pair/direct-site data, traces and provenance. `qc=true` respects the explicit `with_mad` setting. Disabling MAD does not disable medians required by the robust response algorithm. Unrequested or unavailable statistics are marked unavailable, never filled with zero.

Per-detector processing and mask policy

1. Validate ordered identities and geometry; load the supplied master and optional input BPM once. Capture raw-only evidence during ingestion.
2. Correct each raw flat/bias plane once and retain corrected active images, required raw planes/validity, identities and row traces in one owned detector cube.
3. If a BPM is supplied, skip creation and log `input_bpm_supplied`. Otherwise fit a candidate from borrowed corrected flats and publish its FITS/QC.
4. Measure ROI, pair and direct-site data from that same cube. Release it before proceeding to the next detector; assemble newly created BPMs afterward.

The admitted input mask stays fixed; a generated candidate never feeds back into this run's measurement populations. Explicit omission removes only the current mask, preserving historical master/archive lineage. ROI populations can increase; previously admitted direct values must remain exact and newly admitted sites are reported separately. Whole-file preservation hashing still reads inputs before and after the batch; numerical ingestion/correction alone is read-once.

Correction and measured quantities

For frame f, row y, sorted enabled scan reads $v_{f,y,(k)}$, and rejection counts l, h, the existing arithmetic is

$$O_f(y) = \mathrm{pix}\left[\frac{\sum_{k=l+1}^{n-h} v_{f,y,(k)}}{n-l-h}\right], \tag{1}$$

$$U_f(x,y) = \mathrm{pix}[D_f(x,y) - O_f(y)], \qquad S_f(p) = \mathrm{pix}[U_f(p) - B(p)] \quad \text{(flats only)}. \tag{2}$$

Scan summation/division use double precision; pix converts to native `pixelvalue`. Preserve the separate subtraction roundings. Trim is FITS 1-based inclusive; active row j maps to raw row $j + y_{\min} - 1$. Each raw row uses its own scan reads, without smoothing or fitting. The master is already scan-corrected/active and receives no further scan correction. Bias-role frames have no master subtraction. Active BPMs do not mask electronic scan reads.

For each ROI, the common population $\mathcal{P}$ passes the fixed mask and finite/raw-saturation checks in both flats and, when supplied, both raw reference biases. Each member retains its own exposure time, identity and measurement. For $N = |\mathcal{P}|$:

$$\bar{S}_i = N^{-1} \sum_{p\in\mathcal{P}} S_{ip}, \qquad s_i^2 = \frac{\sum_{p\in\mathcal{P}} (S_{ip} - \bar{S}_i)^2}{N-1}, \tag{3}$$

$$d_p = \mathrm{pix}[S_{1p} - S_{2p}], \qquad s_d^2 = \frac{\sum_{p\in\mathcal{P}} (d_p - \bar{d})^2}{N-1}. \tag{4}$$

Corrected counts must match admitted raw counts. For $N < 2$, sample scatter is unavailable. Save member variances, difference variance and $s_d^2/2$ distinctly. Spatial variance is not automatically a noise weight or conversion-gain estimate. Direct sites retain physical pixel values; ROI means do not replace them in a nonlinear pixel model. Units remain ADU, ADU^2 and seconds.

With raw BIAS evidence, raw-pair diagnostics retain their original stage and populations:

$$\mathrm{RON}_{\mathrm{ADU}} = \sqrt{v(B_1 - B_2)/2}, \tag{5}$$

$$\mathrm{RON}_{\mathrm{MAD}} = \frac{1.4826}{\sqrt{2}} \,\mathrm{median}\,|\Delta B - \mathrm{median}\,\Delta B|, \tag{6}$$

$$g = \frac{\bar{F}_1 + \bar{F}_2 - \bar{B}_1 - \bar{B}_2}{v(F_1 - F_2) - v(B_1 - B_2)} \quad [\text{electrons/ADU}]. \tag{7}$$

Without raw BIAS evidence, RON/gain diagnostics are explicitly unavailable; corrected ROI/pair/site measurements continue on the two-flat population. Here v is ordinary sample variance on the declared raw population; MAD is optional. Gain requires a positive denominator and suitable flat pairs; it is a recorded diagnostic, not a conversion applied to the cube. Corrected pair-difference variance must not be passed through this formula with a second bias subtraction. Shared-master variance/covariance are unavailable; no physical fit weights or uncertainties are invented from spatial scatter.

Robust response BPM

For corrected flat j, the reference level x_j is the existing native median of finite eligible pixels in the union of reference ROIs. Prior-mask selection controls the reference population, not whether a pixel is evaluated for defects. Per-pixel valid/unsaturated samples fit the existing Eclipse robust line:

$$S_{pj} \simeq \alpha_p + \beta_p x_j, \qquad A_p = N_p^{-1} \sum_j |S_{pj} - \alpha_p - \beta_p x_j|, \tag{8}$$

$$R_p = \mathrm{pix}\left[\beta_p / \,\mathrm{mean}_{q\in G}\,\beta_q\right]. \tag{9}$$

G contains valid reference slopes. Preserve the fitter's median conventions, native rounding and flagged least-squares fallback. A_p is mean absolute residual, not MAD. Unavailable fits have separate coverage states.

Flags, suitability and products

Recipe `response_low/high` set the current native float32 thresholds: $R_p \leq$ low sets bit value 4; $R_p \geq$ high sets value 8. The unflagged interval is strictly inside 0.5–1.5 by default. These are *dimensionless relative-response limits*, not electrons/ADU. The input master's original configuration remains unchanged; save the current effective thresholds separately in provenance.

The unsigned-byte registry defines the following meanings; the current response fitter creates only values 4/8. Other tests require their own reviewed criteria.

Value	Meaning	Value	Meaning
1	Bias offset	16	Nonlinear response
2	Dark current	32	Temporal noise
4	Low/dead response	64	Instability
8	Excess response	128	Premature saturation

Bits combine by OR. Zero means no detected reason, not certification. Coverage is separate: 0 unavailable, 1 fitted, 2 flagged fallback. `new` output contains newly measured reasons; `combined` requires a known reason-mask prior. A binary prior cannot supply unknown historical reason bits.

Raw saturation and `upper_fraction` limit usable samples; the upper margin and minimum signal span are different controls. Quick mode requires at least five usable pairs with five distinct times/paired levels; provisional mode requires two levels. Save sample counts, signal coverage and pair mismatch. For positive pair reference levels x_1, x_2, the saved mismatch is $|x_1 - x_2|/[(x_1 + x_2)/2]$; compare it with `maximum_pair_difference`. The shared-cube path records failed suitability and continues measuring; it does not relax thresholds, erase evidence or declare a scientifically accepted BPM.

Product	Contents
Detector BPM FITS	Reason image; `COVERAGE`, `NSAMPLE`, `NEWFLAGS`, `SELECTION`, optional `PRIORSEL`.
Detector ASDF	BPM QC, reason/coverage counts, settings, input/HDU mapping and calibrations; linearity measurement ASDF retains ROI/pair/site data and traces.
Final BPM MEF/ASDF	Only newly created BPMs; none for an all-supplied-mask run.
Logs and `native-recipe.toml`	Stage/skip/failure messages, effective recipe, input preservation and truthful completion. Failed work is retained.

BPM image products use `CMDP.<recipe-start>.001.fits` with matching ASDF/log names; existing destinations are refused. A zero exit, completion receipt and independent saved-product checks are required. Saved readers and viewers use measured ASDF values without raw-pixel access or refitting. Separate exploratory residual analysis and future shutter/linearity model specifications remain in their separate guides linked from `docs/developer/README.md`; their eventual consolidation is deferred. They are not steps in this measurement-only recipe. Physical correction and cross-talk remain deferred.

Source map

Entry point: `mefpp2/ins/mef/recipes/mefpp_bpm_linearity.c`. Helpers below are in `mefpp2/ins/mef/common/`.

Stage	Source / landmark
Admission	`mef_bpm_linearity_compile.c`, `mef_project.c`; config/inputs/bindings helpers; `BPM-LINEARITY-ADMISSION`.
Detector/cube	`mef_bpm_linearity_process.c`, `mef_detector_cube.c`; `BPM-LINEARITY-INGESTION`.
BPM / measurements	`mef_cal_bpm.c`, `mef_linearity_pairs.c`, `mef_linearity_prepare.c`; `BPM-LINEARITY-CANDIDATE`, `BPM-LINEARITY-MEASUREMENTS`.
Publication	`mef_bpm_linearity_products.c`, shared native ASDF writer; `BPM-LINEARITY-PUBLICATION`, `BPM-LINEARITY-PRESERVATION`.

Robust arithmetic: `mefpp2/src/iproc/detector_response.c` and `mefpp2/src/math/fit_curve.c`. Scan arithmetic: `mef_overscan_correction_traced`; statistics: `image_pair_stats_checked`.